

# Demonstration of A Figma Plugin to Simulate A Large-Scale Network for Prototyping Social Systems

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## ABSTRACT

Prototyping novel social computing systems is a challenge in the field of Social Computing. Rapid experimentation with novel social network sites can offer valuable insights into their pro-social benefits before their public release to a large audience. In this demo, we present SocialSketch, a Figma Plugin to simulate a crowd in a social network. This demo introduces a plugin for Figma, a no-code interactive prototyping tool, enabling the creation of profile frames and prototype links for large crowds based on realistic network models. Privacy-protective profile content is generated using AI. The plugin aids UX designers, researchers, and students in prototyping social apps and exploring social system design.

## CCS CONCEPTS

• Human-centered computing → Collaborative and social computing.

## KEYWORDS

social computing, prototyping, network simulation, generative AI

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## 1 INTRODUCTION

Billions of users worldwide are connected to each other on large-scale social platforms such as Facebook and Twitter. This massive-scale connectivity has benefits as well as negative consequences such as privacy concerns, mental health, and political polarization.

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Enabling the rapid experimentation of novel social network sites, can allow for valuable insights about pro-social benefits before a public release to a large audience.

Experimenting with the design of novel online social systems is a central challenge of Social Computing research [1, 2, 10, 11]. New design explorations of social systems, particularly large-scale systems Social Network Sites (SNS) that require many users to demonstrate their value, face the extreme hurdles of building a custom infrastructure [9] and obtaining a critical mass of users [1]. Without being able to release a system to a large number of users, it is impossible to predict unintended uses, norms, and other socially emergent adaptations of the intended design [15].

In this demo, we describe a plugin for a popular no-code interactive prototyping tool, Figma, to assist in prototyping a large crowd for social apps. After selecting a type of social network, the plugin produces profile frames on the Figma canvas with prototype links to connect individuals. This creates a crowd of agents using data based on a realistic network model and privacy-protective profile content using generative AI. This plugin can be used by UX designers creating new social networking apps, researchers exploring new design ideas for social systems, and students learning about social computing.

## 2 BACKGROUND

Prototyping is a central phase of the User-Centered Design (UCD) process that allows a design concept to materialize in its interactive form, in order to obtain user feedback for continued cycles of refinement. Low-fidelity prototypes are early-stage prototypes that get refined until becoming a high-fidelity system [12]. A shallow representation of a system might serve the purpose of describing the context of use, whereas a deep and vertical representation might contain fully functional components.

Social affordances, the link between object properties and social characteristics for group interactions [4], are pivotal design dimensions in social computing systems that are difficult to prototype using standard UCD methods. Evaluating a system in its social context requires a significant socio-technical infrastructure that can be prohibitive to rapid prototyping because they require many people to evaluate a system, they evolve over time, they are often unexpected, and people can impact each other in profound

ways [3]. Piggyback prototyping is a social computing evaluation method using existing social platforms to prototype new ones on top of their network to gauge critical mass [9]. This method has privacy and technical drawbacks that can make it prohibitive [7].

Simulations of crowds and social data can provide an avenue for overcoming the challenges of prototyping social affordances [5, 6, 8, 13, 15, 17]. In this demo, we present a tool for simulating a crowd in a large-scale network on Figma, since experiencing an audience is crucial for social platform engagement [1, 2, 14].

### 3 THE SOCIALSKETCH FIGMA PLUGIN

SocialSketch is a Figma Plugin intended for designers and researchers to create new social systems design concepts. Designers and researchers can use the plugin to produce a network of profiles, automatically linked to each other, mimicking a realistic network structure and privacy-preserving profile data.

#### 3.1 Using SocialSketch

When a designer runs the plugin, a list of different types of social networks is available for selection. The designer selects the social network model that best fits their design ideas. Then, the plugin retrieves the corresponding agent-based community from our back-end API and renders the Figma canvas accordingly. Different communities have different numbers of agents, the plugin will generate from 100 up to 400 profiles depending on the selected network. If the designer selected a "networking" community, then each profile frame will be linked to a "mutuals" frame that contains a list of friends. Names listed on the "mutuals" frames are linked to the corresponding agents with prototype connectors. This allows the designer to traverse through the network by clicking through the profiles. When a designer selects the "prototype" mode on Figma, they will see all the prototype links already created to simulated the connections in the network. When a prototype is launched, it is already fully interactive and supports graph traversal by a user. Other networks are based on a 'matching' structure in which profiles are linked to each other through a loop, where the order of the profiles could be based on a matching algorithm and presented in an ordered fashion to a user.

Each profile, and the "mutuals" frame for graph-traversal networks, are instances of two template component frames. The two top-level components are also present on the canvas, a designer can modify these to apply their design simultaneously across all profiles. The crowd produced by SocialSketch can be incorporated into a pre-existing Figma project, for example if a designer already started drafting other pages such as a newsfeed of content, accounts page, or collections of information, using SocialSketch can immediately add 100 to 400 (depending on the selection) user profiles.

#### 3.2 Figma plugin implementation

The Figma plugin was implemented in Typescript, using Node.js and the Figma API. A UI modal implemented in HTML and Javascript provides the list of available network models for selection. After selecting a network, the plugin interfaces with our backend community API to obtain the network data for the user's selection and creates frame components for the profiles and individuals instances of these components with the specific data for each agent. The

plugin also creates a prototype link between an agent's name in a friend's list and its profile frame in the network mode; and a link between the next button on a profile and the next best match in the matching mode.

### 3.3 Agent-based network

**3.3.1 Network modeling for the graph.** The backend of SocialSketch is based on Hashkat [16], a specialized tool for modeling online social communities, incorporating ten variables: network size, geographical region, language, agent personal and ideological interests, and the presence/absence of special agents (celebrities, companies, government entities, bots, political or humorous agents). Using this information, Hashkat then generates a customized network of social media users and their connections, representing their following/follower relationships. To reduce decision-fatigue for a designer, and streamline our prototype, we reviewed all of the Hashkat options and used them to model six network personas based on existing categories of social network and social media sites. These network personas are each backed by a Hashkat output customized to their network parameters.

One network persona mimics large social media networks (e.g., Instagram, Facebook, or Twitter), and has the largest network size of our communities and a very diverse user base with every type of agent present. Another mimics platforms which host smaller communities based on personal connections (e.g., Snapchat or BeReal), and has a small network size, and has a much more homogeneous user base and interests. Another is based on political platforms (e.g., Truth Social or Parler), and has a small network size and homogeneous user base within the same area since politically-focused platforms tend to be highly localized to single country or region. Another network persona mimics interest-based, blog-like platforms (e.g., Tumblr, Pinterest, or Reddit) and has a small network size but a global user base with similar interests and without celebrities, companies, or governments. Another mimics a professional networking platform (e.g., LinkedIn) and has a medium network size and diverse global user base, with different personal and ideological interests. Finally, another network persona is based on community/personality-focused platform (e.g., Twitch, Patreon, or OnlyFans) and has a small network size and a global user base who speak the same language.

**3.3.2 Generative AI for profiles.** Using a network generated by Hashkat as the backbone network for the prototype only produces agent type (Standard and the 5 special agents types) and network information, such as follower and following lists, but not personal information. We then match each agent to a follower/following list of other agents based on their agent type, where certain types such as celebrity, political, and humorous agents having a higher chance to get assigned a higher following count. Each agent was assigned a gender, pronouns, and age from an age range, then to create believable, cohesive user profiles. GPT-3, an AI text generation tool created by OpenAI, was used to generate their name, interests, social media bio, some posts they would write, and their profile picture. Using data generated by an LLMs can result in biased or questionable output [15]. In the case of our Figma plugin, we aimed to broaden representation in the profiles generated by seeding

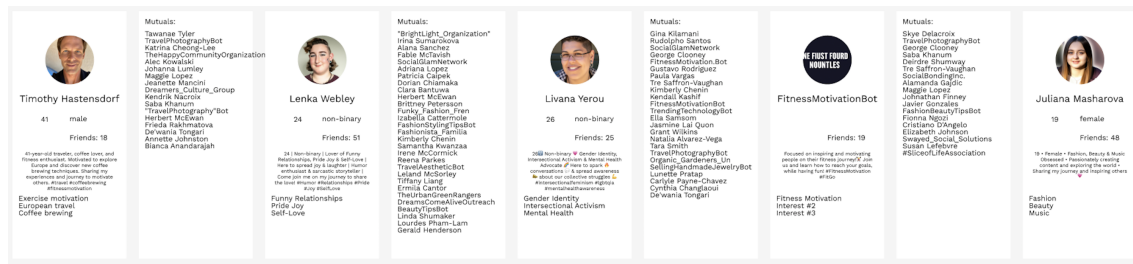


Figure 1: Example profiles and their respective list of friends produced by SocialSketch. This is a subset of the entire crowd.

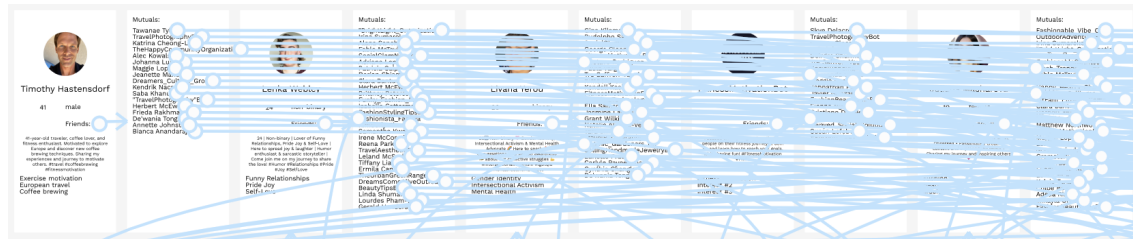


Figure 2: Four example profiles and their respective list of friends, where the names of each friend is linked by a blue connection to its corresponding profile, produced by SocialSketch

a wide range of gender, pronouns, and age. This information is present on each profile on Figma.

## 4 CONCLUSION

Through this demo, we showcase the value of a rapid prototyping tool for generating a crowd in a no-code rapid prototyping tool, Figma. This crowd can be incorporated into an existing Figma project to prototype a new social system, and demonstrate how users will present themselves on the platform. This can be used for evaluating design ideas around self-presentation and audience visibility in social systems. Our future goals include evaluating SocialSketch as a prototyping and creativity tool for envisioning new social systems; and analyzing the diversity, believability, trust, and accuracy of the crowd and agents produced by SocialSketch.

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